

Securing Vehicular Ad Hoc Networks via Digital Twin-Driven Anomaly Detection and Graph Reinforcement Learning

Latifa El Bouga, Amine Andam, Jamal Bentahar, El Mehdi Amhoud, Mustapha Hedabou

Abstract—Vehicular Ad Hoc Networks (VANETs) support critical safety and coordination functions in Intelligent Transportation Systems by enabling vehicles to exchange real-time information. Applications like collision avoidance, cooperative awareness, and traffic optimization depend on reliable Vehicle-to-Vehicle (V2V) communication. However, VANETs remain vulnerable to security threats such as blackhole attacks, message falsification, and data injection due to their dynamic topology and limited interaction durations. While machine learning-based approaches have been applied to anomaly detection in VANETs, most rely on static features or vehicle-level trust, and lack post-detection control strategies. This paper proposes a real-time VANET security framework implemented within a Digital Twin (DT) environment that mirrors the network operation. The framework integrates time-series-based anomaly detection with post-detection mitigation using Graph Reinforcement Learning (GRL). A transfer-learned temporal model, pre-trained on large public data and fine-tuned on simulation outputs, classifies V2V messages based on dynamic behavioral patterns. While a GRL agent observes the evolving communication graph and learns to prune or preserve links to reduce the impact of malicious or misclassified messages. Unlike existing approaches, the system directly addresses false positives and false negatives by learning corrective control policies over time. We implement this framework using OMNeT++, SUMO, and Veins, and evaluate its performance across several road topologies and attack scenarios. Results show improved resilience and reliability compared to detection-only baselines. An ablation study further highlights the value of GRL in enhancing system performance under challenging conditions, achieving a 3.2% improvement in F1-score over a state-of-the-art anomaly detection baseline.

Index Terms—Vehicular Ad Hoc Networks (VANETs), anomaly detection, Digital Twin, time-series analysis, Graph Reinforcement Learning, post-detection mitigation.

I. INTRODUCTION

Vehicular Ad Hoc Networks (VANETs) [1] form a critical component of modern Intelligent Transportation Systems (ITS), enabling real-time exchange of safety and traffic information between vehicles and infrastructure [2]. Applications such as collision avoidance, cooperative lane change, and traffic control rely heavily on accurate and timely Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication [3]. However, the open and decentralized nature of VANETs, characterized by open wireless channels, high node mobility, and the absence of centralized control, makes

them vulnerable to a range of security threats [4], including blackhole attacks, denial-of-service (DoS), Sybil attacks, and false data injection [5], [6].

Ensuring secure message exchange in VANETs is particularly challenging due to their dynamic topology, frequent node mobility, and the absence of persistent trust relationships. Traditional cryptographic techniques and static rule-based models are often insufficient in these environments [7]. As a result, learning-based methods, especially those leveraging machine learning (ML) and deep learning (DL), have been widely studied to detect misbehavior in VANETs [8]. Recent work has also explored ensemble learning approaches for fault detection in VANETs [9], primarily targeting general vehicular faults rather than real-time message-level anomaly handling. However, most existing approaches operate solely at the detection level, using static features or without any mechanism to respond to misclassified or malicious messages during operation. Moreover, many systems assess vehicle-level trustworthiness based on behavioral history (entity-oriented), which can be unreliable in high-mobility environments where interactions are brief and inconsistent [10].

To address this, recent studies have explored data-oriented anomaly detection methods that analyze the real-time content of each V2V message, independent of vehicle history. In particular, time-series models such as LSTMs [11], Transformers [12], and CNN-based architectures have shown success in learning dynamic communication patterns and detecting anomalous message flows [13]. However, these systems typically operate as passive detectors, flagging anomalous messages but providing no mechanism to control their spread or influence on the network.

In parallel, graph-based learning methods, such as Graph Neural Networks (GNNs) and Reinforcement Learning (RL), have gained attention for modeling the structure and evolution of vehicular networks [14], [15]. These techniques allow the system to reason over communication topology, identify structural vulnerabilities, and even guide decisions about message forwarding and routing [16], [17]. Yet, most graph-based systems operate on static graph snapshots, focus exclusively on node-level or routing-level inference, and are rarely coupled with real-time anomaly detection systems [13].

This paper addresses these limitations by introducing a Digital Twin (DT)-based VANET security framework that operates in real time. The proposed system mirrors the VANET simulation environment and integrates time-series-based message classification with graph-level post-detection mitigation. A fine-tuned state-of-the-art temporal model performs message-level anomaly detection, while a Graph Reinforcement Learning (GRL) agent processes the evolving communication graph

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to selectively prune or preserve links. This approach not only mitigates the propagation of malicious data, but also helps correct false positives and false negatives, increasing overall decision reliability. The main contributions of our work are summarized as follows:

- We propose a real-time DT framework for VANET security that unifies anomaly detection and post-detection mitigation in a closed-loop simulation.
- We propose a two-stage training strategy that first trains time-series models on a large-scale public dataset, then fine-tunes them on a smaller, domain-shifted simulation dataset. This approach balances generalization and resource efficiency, enabling real-time classification without requiring massive amounts of simulation data.
- We design a GRL based mitigation strategy that learns to prune or preserve communication links in the message graph, helping correct false positives and false negatives in classification.
- We conducted a thorough ablation study to assess the impact of GRL on system performance and robustness, quantifying its added value over standalone classification.
- We evaluated the full framework using realistic VANET simulation tools (OMNeT++, SUMO, Veins) across multiple attack types and traffic topologies.

The remainder of this paper is organized as follows. Section II presents a review of related work on anomaly detection, graph-based learning, and reinforcement learning in VANET security. Section III introduces the proposed framework, including the DT architecture, the anomaly detection module, and the GRL-based control layer. Section IV details the training methodology, covers the datasets, and the model training procedures. Section V describes the experimental setup, test datasets, attack scenarios, evaluation results, and the ablation study. Finally, Section VI discusses limitations and outlines directions for future work, followed by the conclusion in Section VII.

II. RELATED WORK

Security and anomaly detection in VANETs have been widely studied using ML, DL, GNNs, and RL. Prior works generally fall into four main categories: ML/DL-based anomaly detection, GNN-based modeling, RL-based adaptation, and frameworks that combine detection with post-detection control. One key modeling difference lies in whether a system analyzes vehicles as entities (evaluating their long-term behavior) or evaluates messages individually based on their real-time content. The latter, known as data-oriented detection, is often better suited for VANETs, where vehicles frequently join and leave the network and may not retain a persistent history.

A. Anomaly Detection in VANETs Using Machine and Deep Learning

Entity-oriented detection is still common in routing-focused security systems. Hassan et al. [18] proposed an artificial neural network that is trained to identify blackhole attacks based on vehicle behavior, which is then used to support enhanced

routing. Although effective in improving routing security, the model relies on historical data on node trustworthiness and does not support message-level or real-time adaptation.

Canh et al. [19] used features generated from NS-3 simulations [20] and applied classifiers such as Gradient Boosting to detect malicious nodes. Similarly, El-Shafai et al. [21] applied a combination of Random Forest, Extra Trees, and CNN to physical layer metrics such as Received Signal Strength (RSS) and Bit Error Rate (BER). While both systems achieve strong performance, they are limited by not providing post-detection mitigation.

A more data-oriented method is proposed by Alladi et al. [22]. This model uses a CNN-LSTM architecture to learn patterns in communication flows and detect a variety of attack types without requiring labeled data. Although the system analyzes message content and generalizes well, it does not integrate mitigation or support structural changes to communication patterns during operation.

While many anomaly detection methods in VANETs rely on static features extracted from individual messages, similar machine learning-based intrusion detection approaches have also been explored in broader networked and IoT environments [23], highlighting shared challenges related to scalability and real-time adaptation. However, such models often overlook how vehicle behavior unfolds over time. In a dynamic and fast-changing environment like VANETs, threats may manifest as subtle temporal inconsistencies, such as gradually drifting positions or fluctuating speeds that static classifiers fail to detect. Time-series models, on the other hand, are designed to capture sequential patterns and temporal dependencies, making them more effective in identifying evolving attack behavior. Prior work in sequential anomaly detection has shown that such models can outperform traditional approaches [22], [24]. Building on this insight, our framework incorporates fine-tuned temporal models to perform real-time message-level classification. These outputs are then used to inform post-detection mitigation within a DT, enabling both accurate detection and adaptive response.

B. Graph-Based Learning in VANET Security

Graph-based methods have emerged to model relationships between vehicles and communication patterns. Ramkumar et al. [13] use a semantic-aware GNN to detect malicious vehicles by modeling node and edge relationships. However, the approach relies on static graph snapshots and performs only node-level classification, without responding to detected anomalies at the link or message level. In a more comprehensive setup, Hidalgo et al. [14] proposed the SerIoT platform, which combines GNN-based detection with Software Defined Network (SDN) and RL for flow control and rerouting. While promising, the platform is heavy on infrastructure and depends on SDN controllers and fog nodes.

C. Reinforcement Learning in VANET Security

RL has been explored to enable adaptive trust and routing decisions in VANETs. Guo et al. [25] presented a Q-learning agent that dynamically adjusts the weighting between internal

sensor data and external V2V messages when evaluating incoming information. The model is data-oriented, as it assesses each message in its situational context (e.g., event type, location, entropy), rather than relying on historical vehicle behavior. However, it does not model the communication structure as a graph and lacks the ability to enforce structural mitigation, such as pruning or blocking message links.

Sedar et al. [26] proposed a DRL-based system in which RSUs share knowledge across domains to detect misbehavior under unseen conditions. While this enables generalization, the approach focuses on node-level detection without supporting real-time message-level control, or structural adaptation of the communication graph.

D. Beyond Detection: Learning-Based Detection and Post-Detection Response

Some frameworks extend anomaly detection by combining time-series modeling with RL. He et al. [27] presented a time-series anomaly detector for traffic flow data using LSTM to encode temporal patterns and a DQN [28] agent to classify data points as normal or anomalous. In this case, RL is used directly for detection decisions, not for post-detection mitigation, and the system is not designed for VANET environments, unlike our framework, which uses RL for real-time mitigation in a VANET setting.

Golchin et al. [29] proposed a time-series anomaly detection framework that combines LSTM-based modeling with RL and variational autoencoders (VAE). The RL agent learns to detect anomalies by balancing rewards from limited labeled data and unsupervised VAE signals. While this enhances detection, the system is not applied to VANETs and does not perform post-detection control or communication-level decision-making, capabilities that are explicitly addressed in our work.

Other studies apply RL within closed-loop frameworks that include control or response mechanisms. Jayakrishna et al. [30] proposed a VANET security system that detects DDoS attacks using DL and applies RL to adjust the network rate limits. Their control is applied at the infrastructure level (e.g., RSUs), which may impact both malicious and legitimate traffic. The method also focuses only on DDoS attacks and does not address more general forms of message falsification, unlike our system, which operates at the message level and targets a broader range of attacks.

Zhang et al. [31] proposed an autonomous driving system that uses LSTM to predict surrounding vehicle behavior and RL to make lane-change and car-following decisions. Although this combines time-series forecasting and decision-making, it is designed for motion control in autonomous vehicles rather than VANET security or communication-level response.

While these works combine time-series models and RL in useful ways, none of them supports message-level mitigation or uses RL to make post-detection decisions within a VANET communication context, as addressed in our framework.

E. Privacy and Authentication in VANETs

In parallel to detection and mitigation, several works have addressed privacy-preserving communication and secure au-

thentication in VANETs. Tan et al. [32] proposed a blockchain-assisted conditional anonymous authentication protocol with adaptive group key agreement. Their system ensures identity privacy, revocation, and secure cross-domain communication using smart contracts and tree-based key structures. Similarly, Ismail et al. [33] introduced a biometric-enhanced anonymous key agreement scheme that leverages smart cards, fuzzy extractors, and elliptic curve cryptography to provide fine-grained access control and conditional traceability. While our framework focuses on detection and mitigation rather than communication-layer privacy or authentication, it is compatible with existing privacy-preserving protocols and can be extended to operate alongside them in future deployments.

In contrast, our work presents a complete, time-series-driven security framework that spans the entire security pipeline, from message generation and anomaly detection to post-detection control. The system is implemented within a DT that mirrors the VANET simulation in real time, enabling synchronized classification and mitigation. It performs message-level anomaly detection using a temporal DL model and applies GRL to dynamically prune or preserve links in the communication graph. The GRL agent learns from previous predictions and local graph context, helping to reduce false positives and false negatives while maintaining communication integrity. Together, these components form a unified real-time system for detection and mitigation in VANETs.

III. PROPOSED METHODOLOGY

A. Overview of the Proposed Approach

This work introduces a real-time DT framework designed to enhance VANET security through integrated detection and mitigation of malicious V2V communications. The DT mirrors the behavior of a simulated VANET by continuously receiving live vehicle state data and performing real-time analysis through two core modules (see Figure 1): a time-series anomaly detection module and a GRL agent module for post-detection control.

The VANET environment is emulated using SUMO (Simulation of Urban MObility) [34] for mobility modeling and OMNeT++ (Objective Modular Network Testbed in C++) [35] with Veins (Vehicles in Network Simulation) [36] for wireless communication. Vehicles periodically exchange structured V2V messages that contain movement-related attributes such as position, speed, acceleration, and heading. These messages are streamed in real time to the DT back-end, enabling synchronized monitoring and decision-making.

Within the DT, the first module uses a fine-tuned time-series classification model to analyze sequences of recent vehicle states and classify each message as normal or anomalous. These predictions are then fused with vehicle metadata to construct a dynamic VANET communication graph, where nodes represent vehicles, and edges represent active V2V communication links. The node features include physical movement data and binary anomaly labels from the classification model.

The second module of the DT applies GRL to this constructed graph. The problem is formulated as a Markov Decision Process (MDP), where the agent observes the graph

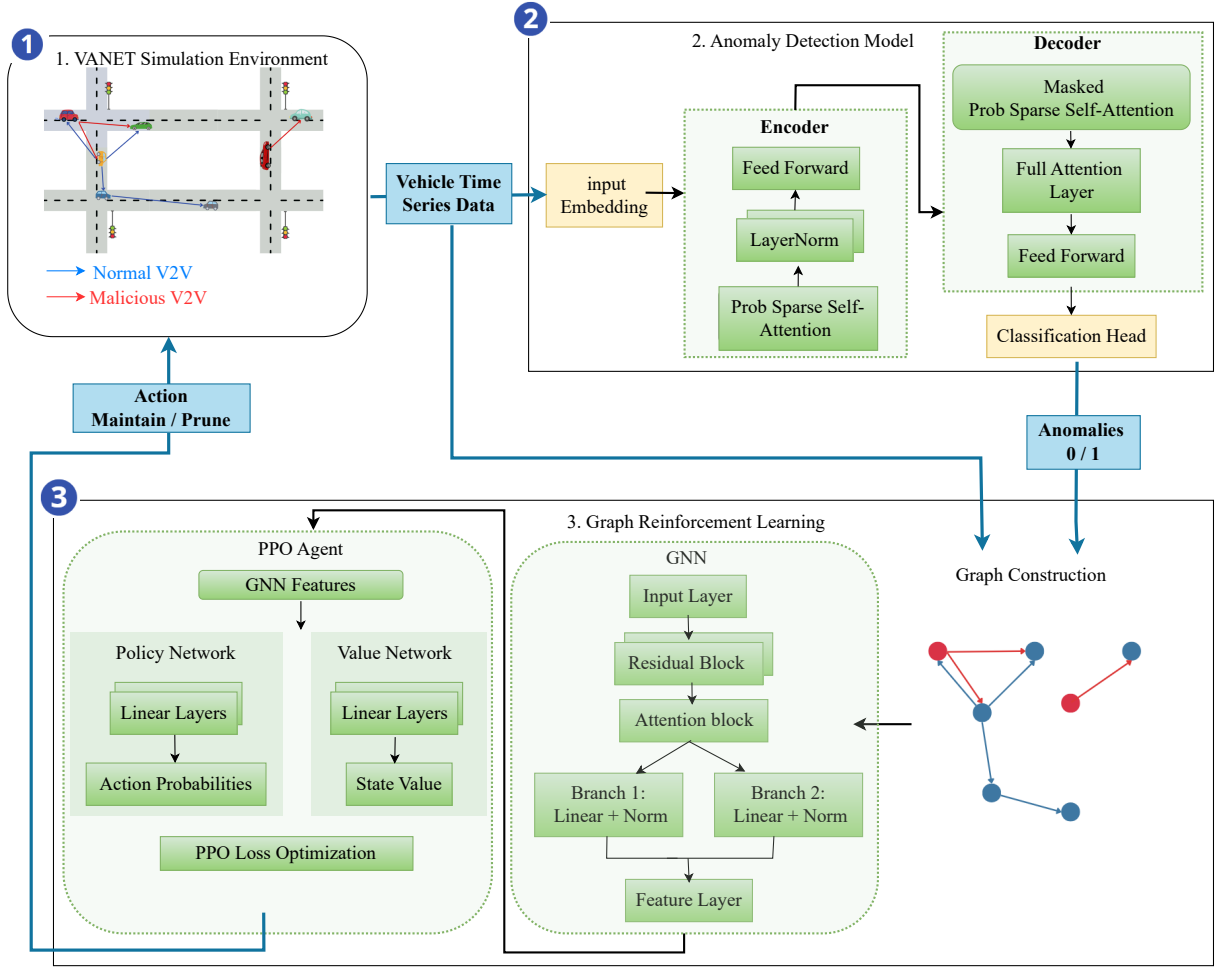


Fig. 1. Overview of the proposed VANET security framework. The system operates as a simulation-driven Digital Twin that integrates time-series anomaly detection model and Graph Reinforcement Learning for real-time classification and mitigation of malicious V2V messages.

at each time step and decides, for each communication link, whether it should be maintained or pruned. A GNN extracts latent node features, which are processed by a Proximal Policy Optimization (PPO) [37] policy to generate edge-level actions. The reward function is designed to balance detection precision, ensuring that the agent suppresses malicious communication links while minimizing disruption to normal traffic flow.

Figure 1 summarizes the overall system architecture. The simulation environment (Block 1) generates live vehicle messages, which are transmitted to the anomaly detection module (Block 2) for real-time classification. The resulting predictions, combined with vehicle state information, are used to construct a dynamic communication graph. This graph is then processed by the control module (Block 3), where a GNN–PPO policy evaluates each communication link and decides whether to prune or retain it. These control actions are sent back to the simulation in real time, forming a closed-loop DT that can both monitor and influence network behavior securely.

With the overall architecture defined, we now detail the two core components of the proposed DT framework. We begin with the anomaly classification module, which detects message-level anomalies based on temporal behavior patterns in vehicle communication.

B. Anomaly Classification module

The anomaly classification module represents the first stage of the DT framework, responsible for detecting abnormal V2V communication behavior in real time. Operating continuously alongside the simulated VANET, this module monitors vehicle activity and produces binary predictions that inform the downstream decision-making process.

We formulate anomaly detection as a binary time-series classification task. Each V2V message m_t generated at time t , is associated with a temporal sequence of past observations:

$$\mathbf{X}_t = [\mathbf{x}_{t-T+1}, \mathbf{x}_{t-T+2}, \dots, \mathbf{x}_t] \in \mathbb{R}^{T \times d}. \quad (1)$$

where $\mathbf{x}_t \in \mathbb{R}^d$ denotes the state vector of the vehicle at time t , T is the size of the temporal window, and d is the dimensionality of the features. Each $\mathbf{x}_t$ includes a set of physical and contextual features:

$$\mathbf{x}_t = [\text{PosX}_t, \text{PosY}_t, \text{SpdX}_t, \text{SpdY}_t, \text{AcI}_t, \text{AcIY}_t, \text{HedX}_t, \text{HedY}_t] \quad (2)$$

The objective is to learn a classification function f_θ , parameterized by model weights θ , that maps each input sequence

$\mathbf{X}_t$ to a binary output:

$$\hat{y}_t = f_\theta(\mathbf{X}_t) \in \{0, 1\} \quad (3)$$

where $\hat{y}_t = 1$ denotes a malicious message and $\hat{y}_t = 0$ indicates a normal one.

During training, the model is supervised using ground-truth labels $y_t \in \{0, 1\}$, which indicate whether the message at time t is actually malicious. The model parameters θ are optimized using the Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{\text{BCE}} = - \sum_t [y_t \log(\hat{y}_t) + (1 - y_t) \log(1 - \hat{y}_t)] \quad (4)$$

To implement this classification function, we evaluate several state-of-the-art deep temporal models known for their effectiveness on sequential data:

- **TimesNet** [38] a convolutional neural network that captures long-range temporal patterns using temporal convolutional blocks.
- **Non-Stationary Transformer (NST)**, [39] a Transformer-based model designed to handle evolving temporal distributions.
- **Informer**, [40] another Transformer-based model optimized for efficient modeling of long sequences through probabilistic self-attention mechanisms.

Each of these models takes the sequence $\mathbf{X}_t$ as input and outputs a probability indicating the likelihood of a malicious message. The final binary prediction $\hat{y}_t$ is then used as an input feature in the graph constructed by the next module GRL. This allows the agent to reason over both physical dynamics and inferred message trustworthiness.

While this module provides fast, data-driven detection, it may still produce false positives FP (incorrectly flagging normal messages) and false negatives FN (missing actual attacks). These uncertainties motivate the use of the GRL module, which refines the final security decisions by evaluating predictions in the context of the overall communication graph.

C. Graph Reinforcement Learning module

While the anomaly classifier produces real-time, message-level predictions, its output may contain errors such as FP or FN. To enhance robustness, the DT framework includes a GRL agent. This agent operates on a dynamic graph representation of the VANET and learns to selectively prune unsafe communication by observing the evolving structure and behavior of the network.

We formulate this control task as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where:

1) **State Space** $\mathcal{S}$: At each timestep t , the VANET is represented as a directed graph $G_t = (V_t, E_t)$, where:

- V_t is the set of vehicles (nodes),
- $E_t \subseteq V_t \times V_t$ is the set of active V2V communication links (edges).

Each edge $e_{ij} \in E_t$ is encoded as a feature vector:

$$s_{ij}^t = [\text{SenderID}_i, \text{ReceiverID}_j, \text{Timestamp}, \text{PosX}, \text{PosY}, \text{SpdX}, \text{SpdY}, \text{AclX}, \text{AclY}, \text{HedX}, \text{HedY}, \text{AnomalyLabel}, \text{FP Rate}, \text{FN Rate}] \quad (5)$$

Each s_{ij}^t includes:

- **Physical state** of the sender vehicle (position, speed, acceleration, heading).
- **Anomaly Label** predicted by the anomaly detection model indicating whether the message is predicted to be malicious.
- **Communication Metadata**: The message timestamp and the sender and receiver indices, identifying the link context.
- **Temporal Risk Statistics**: The FP and FN rates, computed over a sliding window to capture recent classification trends and guide the GRL agent's response to evolving uncertainty.

2) **Action Space** $\mathcal{A}$: The GRL agent must decide, for each edge $e_{ij} \in E_t$, whether to allow or block communication:

$$a_{ij}^t \in \{0, 1\}, \quad \text{where } 1 = \text{Prune}, 0 = \text{Maintain} \quad (6)$$

The action vector A_t includes a binary decision for every edge in the graph. These actions are applied at the end of each timestep, and communication on pruned links is blocked in the subsequent simulation step.

3) **Reward Function** $\mathcal{R}$: Unlike traditional environments where each time step has a scalar reward, our agent receives per-edge rewards based on the correctness of each pruning decision. The reward function balances True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN). For each edge e_{ij} , the reward r_{ij}^t is computed as:

$$r_{ij}^t = \begin{cases} +\alpha \cdot w_{\text{mal}} & \text{if } a_{ij}^t = 1 \wedge y_{ij} = 1 \quad (\text{TP}) \\ -\beta \cdot w_{\text{norm}} & \text{if } a_{ij}^t = 1 \wedge y_{ij} = 0 \quad (\text{FP}) \\ -\gamma \cdot w_{\text{mal}} & \text{if } a_{ij}^t = 0 \wedge y_{ij} = 1 \quad (\text{FN}) \\ +\delta \cdot w_{\text{norm}} & \text{if } a_{ij}^t = 0 \wedge y_{ij} = 0 \quad (\text{TN}) \end{cases} \quad (7)$$

where:

- a_{ij}^t : Agent's action (1 = prune, 0 = keep)
- y_{ij} : Ground-truth class (1 = malicious, 0 = normal)
- $w_{\text{mal}}, w_{\text{norm}}$: dynamic class weights based on the proportion of malicious and normal messages in the current timestep, calculated as:

$$w_{\text{mal}} = \frac{N_{\text{mal}}}{N}, w_{\text{norm}} = \frac{N_{\text{norm}}}{N} \quad (8)$$

where N_{mal} is the number of malicious messages (i.e., messages with $y_{ij} = 1$ at the current timestep, N_{norm} is the number of normal messages (i.e., messages with $y_{ij} = 0$, and $N = N_{\text{mal}} + N_{\text{norm}}$. These weights emphasize the minority class by giving it proportionally higher influence in the reward function.

To prevent reward accumulation from masking poor decisions (e.g., being rewarded for pruning some malicious links while still maintaining others), we implement adaptive penalty scaling. The environment tracks recent FP and FN rates over a sliding window and increases β or γ dynamically when the corresponding error rate increases. This adaptive shaping ensures that the agent focuses on improving its weakest decision type over time. The agent's total reward at time t is the sum of edge-level rewards:

$$R_t = \sum_{e_{ij} \in E_t} r_{ij}^t \quad (9)$$

The agent learns a policy π_θ that maximizes the expected cumulative discounted reward.

$$\pi^* = \arg \max_{\pi} E_{\pi} [G_t] = \arg \max_{\pi} E_{\pi} \left[\sum_{t=0}^T \gamma^t R_t \right] \quad (10)$$

where $\gamma \in [0, 1]$ is the discount factor.

4) **GRL Agent Architecture:** The GRL policy network is composed of a GNN used as a feature extractor, followed by two output heads trained using PPO. The GNN processes graph-structured state information and encodes it into a compact latent representation that supports link-level decision-making.

The architecture, see figure 1, begins with a fully connected layer, followed by two residual blocks with batch normalization, gated skip connections, and dropout. This is followed by a self-attention block to model long-range interactions. And a multi-branch downsampling unit to compress node embeddings into a shared feature vector.

This representation is shared by two separate heads:

- The policy head: produces edge-level logits that are thresholded into binary actions (prune or maintain).
- The value head: estimates the scalar value of the current graph state, providing a baseline for policy optimization.

5) **Training Strategy:** The GRL agent is trained offline using PPO on data collected from simulated VANET trajectories. At each training step, a graph is constructed from the vehicle states and the predictions. Rewards are computed using the adaptive shaping strategy, and gradients are optimized using the clipped PPO objective.

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip} \left(r_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) + c_1 \cdot \text{MSE}(V(s_t), G_t) - c_2 \cdot \mathcal{H}[\pi_\theta] \right] \quad (11)$$

where $r_t(\theta)$ is the ratio between the new and old policy probabilities, $\hat{A}_t$ is the advantage estimate from Generalized Advantage Estimation (GAE) [41], and $\mathcal{H}[\pi_\theta]$ is the entropy of the policy to encourage exploration. The constants c_1 and c_2 weight the value loss and entropy regularization.

Hyperparameters such as batch size, learning rate, and entropy coefficient are tuned for training stability.

6) **GRL Decision-Making Mechanism:** At each decision step, the GRL agent observes a dynamically constructed V2V communication graph, where nodes represent vehicles, and edges correspond to active V2V message exchanges. Each edge is encoded with a feature vector that captures the physical state and communication context, including sender and receiver identifiers, message timestamp, vehicle position, speed, acceleration, heading, the binary anomaly label produced by the detector, and the recent false positive and false negative rates computed over a sliding window. The agent processes the full graph using a GNN encoder and outputs a binary action for each edge: *maintain* (retain the communication link) or *prune* (suppress the link). These decisions are applied after each detection cycle in real time, allowing the communication graph to adapt dynamically based on evolving message behavior. The policy is trained using PPO to maximize a reward that promotes correct suppression of malicious links while preserving legitimate communication.

IV. TRAINING METHODOLOGY

This section presents the full training methodology for the proposed DT-based VANET security framework. We use a combination of a large-scale public dataset (VeReMi) [42] and simulation-generated data to enable both generalization and environment-specific adaptation. The entire training process that covers data, model training, and policy optimization is summarized in Algorithm 1.

Algorithm 1 Training Pipeline for the VANET Digital Twin Framework

Input: VeReMi dataset D_{public} , simulation-generated dataset D_{sim}

Output: Trained anomaly detection model f_θ , trained GRL policy π_ϕ

1. Train Anomaly Detector

Pre-train time-series model f_θ on D_{public}

Fine-tune f_θ on simulation data D_{sim}

2. Generate Graph-Based Training Data

Run simulation and classify messages using f_θ

Construct communication graphs using vehicle state and predicted labels

3. Train GRL Agent

Train GRL policy π_ϕ using PPO on collected graph trajectories

A. Dataset Description

This study uses a combination of public and simulation-generated datasets to train, adapt, and evaluate the components of the proposed VANET security framework. All datasets consist of structured V2V messages that contain vehicle behavioral features such as position, speed, acceleration, heading, sender/receiver IDs, and timestamps. Each message is labeled as normal (0) or malicious (1), depending on whether it was affected by an injected attack.

1) **VeReMi Dataset:** The Vehicular Reference Misbehavior (VeReMi) dataset [42] is a widely used benchmark for anomaly detection in VANETs. It includes multiple attack scenarios such as GPS falsification, message flooding, and Sybil attacks. In this work, VeReMi is used to train and benchmark the anomaly classifier within the DT framework. The dataset includes approximately 3.2 million V2V messages, with around 40.5% labeled as malicious. The class distribution is illustrated in Figure 2.

2) **Simulation-Generated Dataset:** To adapt the anomaly detector to our simulation domain and train the GRL agent, we generated custom datasets using our simulation framework. Two simulation runs were conducted, each with different randomized parameters and attack schedules:

- The first dataset was used to fine-tune the anomaly detector on domain-specific behavioral patterns.
- The second was used to train the GRL agent, enabling it to learn context-aware link pruning strategies

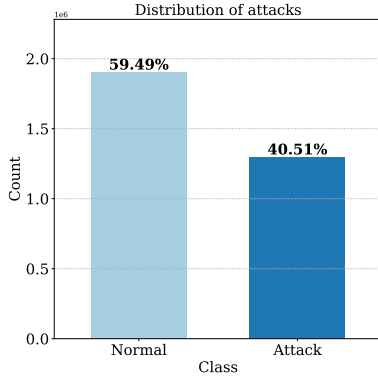


Fig. 2. The class distribution of VeReMi Dataset

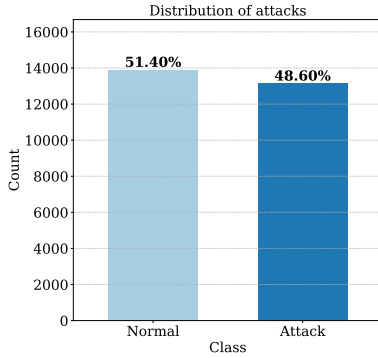


Fig. 3. The class distribution of Generated Dataset

To ensure proper domain adaptation and avoid leakage, the simulation data used for training is completely disjoint from the datasets used during evaluation. The types of attacks used in these training datasets are described in Section V. Figure 3 shows the class distribution of one simulation run used for training.

B. Anomaly Detector Training

To assess the effectiveness of time-series anomaly classification, we benchmark three state-of-the-art deep temporal models on the VeReMi dataset: TimesNet, Non-Stationary Transformer (NST), and Informer. These models are designed to capture temporal dependencies in sequential data, a critical capability for detecting malicious V2V messages in dynamic vehicular environments.

All models were trained using a consistent set of input features, including vehicle position, speed, acceleration, and heading. The dataset was split into 80% training and 20% testing. Training was conducted for a maximum of 20 epochs using the Adam optimizer and early stopping based on validation loss to prevent overfitting.

Figure 4 shows the performance of the training in terms of loss, accuracy and F1-score over epochs. TimesNet showed the fastest convergence and highest training performance, while Informer and NST followed closely with more gradual but stable improvements.

To assess generalization, all models were evaluated on the test set. Table I presents the accuracy and F1-score achieved

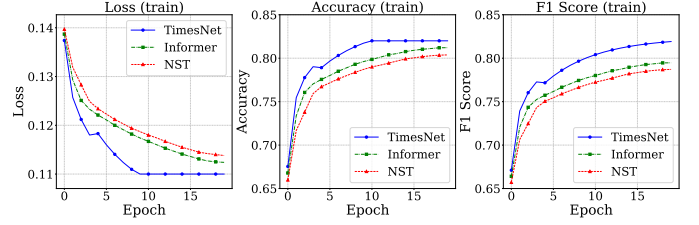


Fig. 4. Training loss, accuracy, and F1-score over epochs for TimesNet, Informer, and NST trained on the VeReMi dataset

TABLE I
ACCURACY AND F1-SCORE OF EACH DT MODEL ON THE VEREMI TEST SET

Model	Accuracy	F1-Score
TimesNet	0.8192	0.8077
Informer	0.8190	0.8067
NST	0.8168	0.8041

TABLE II
ACCURACY AND F1-SCORE OF DT MODELS ON SIMULATION-GENERATED DATA BEFORE FINE-TUNING.

Model	Accuracy	F1-Score
TimesNet	0.4669	0.3743
Informer	0.4555	0.3534
NST	0.4283	0.3199

by each model. All three achieved strong classification performance, with TimesNet slightly outperforming the others in both accuracy and F1-score.

C. Distribution Shift & Fine-Tuning

Although the anomaly detection models performed well on the VeReMi dataset, their performance deteriorated significantly when applied directly to our simulation environment. As discussed in Section IV-A2, we generated new datasets, which introduced variability in road structures, vehicle dynamics, message timings, and randomized attack scheduling. These differences introduced a distribution shift between the training domain (VeReMi) and the target simulation environment used throughout this work.

This shift impacts model generalization. The TimesNet, Informer, and NST models were initially trained on the structured patterns present in VeReMi, which do not fully reflect the diversity and randomness of our simulation-based scenarios. When evaluated on one of our generated datasets, all models exhibited a substantial drop in accuracy and F1-score (see Table II), confirming the need for adaptation.

While one option would be to train anomaly detection models directly on simulation data, this approach presents several limitations. First, our simulation datasets contain only about 27,000 labeled samples, whereas VeReMi offers more than 3 million V2V messages, making them more suitable for training deep temporal models. Second, collecting large amounts of simulation data is both time-consuming and computationally expensive. As a result, training models from scratch on simulation-only data would likely lead to suboptimal performance and a high data collection cost.

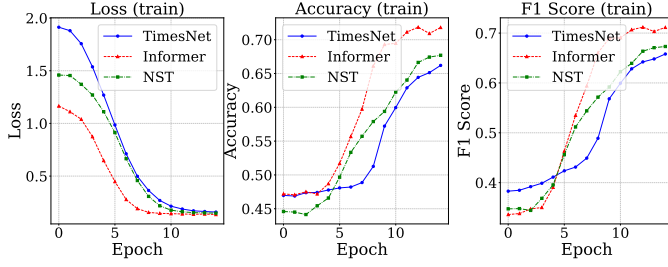


Fig. 5. Training loss, accuracy, and F1-score curves for fine-tuning DT models on simulation data.

TABLE III
POST-FINETUNING ACCURACY, F1-SCORE, GPU MEMORY USAGE, AND INFERENCE TIME.

Model	Accuracy	F1-Score	GPU Mem	Inf. Time
TimesNet	0.6033	0.6136	17.91 GB	228.24 s
Informer	0.8134	0.8249	0.17 GB	33.20 s
NST	0.6199	0.6222	0.19 GB	33.50 s

To overcome this, we adopt a two-stage training strategy: models are first trained on VeReMi to capture general communication patterns, then fine-tuned on simulation output to adapt to scenario-specific dynamics.

Figure 5 shows the progress of the training of all three models during fine-tuning, including loss, accuracy, and the F1-score in epochs. Informer exhibited the most stable and rapid convergence, followed by NST and TimesNet.

After fine-tuning, the models were re-evaluated on the test set. Table III summarizes the results, including classification performance and resource usage. Informer achieved the highest classification performance while also requiring significantly less GPU memory and inference time. Based on these results, Informer was selected as the final anomaly detection model for all remaining experiments. It offers the best balance between classification accuracy and real-time inference efficiency, which makes it well suited for integration into the real-time simulation loop described in Section III-A.

D. Graph Construction

Once the anomaly detection model is fine-tuned, its outputs are used to generate graph-based observations for GRL training. Each node in the graph represents a vehicle and is assigned a feature vector containing physical state information (e.g., position, velocity, acceleration, heading), along with the binary anomaly labels generated by the classifier. Edges represent active V2V communication links between vehicles. These node and edge definitions form a dynamic communication graph, which is constructed at each simulation timestep. The graph is updated every 0.1 seconds (see Table IV), aligned with the beacon interval used in the communication protocol. This frequency ensures that the GRL agent receives up-to-date inputs reflecting recent V2V interactions. It provides a balance between real-time responsiveness and computational efficiency: more frequent updates would increase system latency, while slower updates could delay corrective actions and reduce mitigation effectiveness.

E. GRL Agent Training

The GRL agent is trained offline using the communication graph sequences generated during simulation. At each timestep, the agent observes a graph and outputs edge-level decisions to prune or maintain each communication link. After each decision, a reward is computed for every edge based on the accuracy of the pruning action. The reward formulation, introduced in Section III-C3, balances precision and recall using class-weighted signals and adaptive shaping based on recent trends in false positives and false negatives.

The agent is optimized using the PPO algorithm. The GRL policy network consists of a GNN encoder followed by two output heads: one for the edge-level pruning actions and one for the state value estimation. Advantage estimates are computed using GAE to guide policy updates. PPO was selected due to its state-of-the-art performance reported on a wide range of RL tasks [37]. As a policy gradient method, PPO offers stable learning and is relatively easy to implement, making it well-suited for training in dynamic, graph-based environments such as VANETs. Compared to value-based methods such as DQN, PPO has been shown to perform more reliably in high-dimensional scenarios, which are typical in our message-level mitigation task [43].

V. EXPERIMENTAL SETUP AND RESULTS

We now evaluate the effectiveness of the proposed framework in a simulated VANET environment, using unseen road topologies, varying traffic conditions, and multiple attack scenarios.

A. Simulation Scenario Setup

To emulate realistic vehicle mobility and communication behavior, we implement a simulation-driven DT using a hybrid framework composed of OMNeT++, SUMO, and Veins (a video demonstration is available)¹:

- **OMNeT++** handles the simulation of the discrete event network.
- **SUMO** performs microscopic vehicular traffic simulation.
- **Veins** integrates OMNeT++ and SUMO through TraCI (Traffic Control Interface) [44], enabling real-time interaction between vehicle mobility and communication.

The details of the full simulation environment are summarized in Table IV.

1) *Real-World Roads Topologies*: To ensure environmental diversity and urban realism, three real-world road networks were extracted from OpenStreetMap (OSM) and manually refined using JOSM (Java OpenStreetMap Editor). This pre-processing removed invalid geometries and simplified the networks for simulation.

- **Road 1**: A simpler layout with roundabouts from Marakech, Morocco.
- **Road 2**: A denser and irregular urban zone from Casablanca, Morocco.
- **Road 3**: A structured urban grid from Orlando, USA.

¹A demonstration video link <https://youtu.be/LPpwGdiFBXw>

TABLE IV
SIMULATION ENVIRONMENT DETAILS

Parameter	Value
<i>Simulation Tools</i>	
Traffic Simulator	SUMO 1.8.0
Network Simulator	OMNeT++ 5.6.2
V2X Framework	Veins 5.2
<i>Road Scenarios</i>	
Road 1	2.7 km × 2.1 km
Road 2	12.5 km × 6.9 km
Road 3	6.9 km × 9.0 km
Map Source	OpenStreetMap
<i>Traffic Configuration</i>	
Simulation Time	500 s
Number of Vehicles	20, 50, 100
Vehicle Generation Rate	1 veh/s
Number of Lanes	1–2 per direction
Vehicle Speed	50–100 km/h
<i>Communication Configuration</i>	
MAC Protocol	IEEE 802.11p
Frequency Band	5.9 GHz
Channel Bandwidth	10 MHz
Transmission Power	20 mW
Beacon Interval	0.1 s
<i>Attack Configuration</i>	
Malicious Vehicle Ratio	10%, 20%, 30%, 50%
Attack Types	DoS, Random Speed, Position Offset

These maps provide varying traffic conditions, road connectivity, and topology complexity (see Figure 6).

Each road was simulated with three different vehicle densities (20, 50, 100 vehicles) and four levels of malicious node presence (10%, 20%, 30%, 50%). Vehicles exchange V2V messages within a 400-meter communication radius.

2) *Attack Injection Models*: To evaluate the system under adversarial conditions, we implement three realistic message-level attack models, each designed to degrade the quality of shared information differently:

- **DoS Disruptive Attack**: This attack aims to overload the network by generating a large volume of falsified motion data. It achieves this by injecting speed values that follow a sinusoidal pattern and by synthesizing false acceleration and heading information. The impact of this attack lies in its ability to flood the communication channel and overwhelm receivers with contradictory signals.
- **Random Speed Attack**: The objective of this attack is to subtly corrupt the velocity data in a way that appears plausible to anomaly detectors. It operates by adding Gaussian noise ($\sigma = 15/3$) to the actual speed values, while also introducing small random perturbations to acceleration.
- **Constant Position Offset Attack**: This attack persistently shifts the vehicle's reported position by a fixed spatial offset ($\sigma = 40\text{meters}$), while preserving normal-looking speed and acceleration values. It subtly injects location errors over time, causing the vehicle to appear to be offset from its actual trajectory. This creates “ghost vehicle” effects in the perception of neighboring nodes.

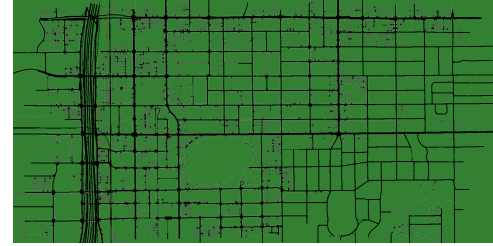
Each attack is triggered probabilistically during the simulation based on the assigned malicious vehicle ratio and may vary in temporal and spatial execution to simulate various real-world threat patterns.



(a)



(b)



(c)

Fig. 6. Real-world road networks used in the simulation experiments. (a) Road 1: Marrakech, Morocco. (b) Road 2: Casablanca, Morocco. (c) Road 3: Orlando, USA.

3) *Hardware Setup*: All simulations were executed on a desktop machine equipped with an Intel Core i7-10700 CPU @ 2.90 GHz and 16 GB RAM. Training for anomaly detection models and the GRL agent was performed offline on a NVIDIA A100-SXM4 GPU with 80 GB memory. All simulation runs were initialized with different random seeds to ensure statistical diversity and reproducibility.

4) *Real-Time System Integration*: The proposed framework integrates a VANET simulation with a Python-based back-end responsible for real-time anomaly detection and post-detection control. This setup forms a closed-loop system where V2V messages are classified in real time, and link-level control actions are enforced dynamically within the simulation. Communication between OMNeT++ and the Python back-end is established using a TCP/IP socket based on the Winsock API. At each simulation timestep, when a vehicle transmits a message, relevant data such as sender ID, position, velocity, acceleration, heading, and timestamp are extracted and sent to the Python server. The server reconstructs the input time-series sequence for the message sender and uses a checkpointed Informer model to classify the message as normal or malicious. The binary prediction, along with the corresponding vehicle state and context, is then embedded in a dynamic communication graph. This graph is passed to the GRL agent,

TABLE V
GNN STRUCTURE

Parameter	Value
Number of Layers	2 Residual Blocks + 1 Attention Layer
Hidden Dimensions	512 $\rightarrow$ 768 $\rightarrow$ 512
Output Dimension	512
Attention Heads	8
Activation Function	GELU
Dropout Rate	0.25
Normalization	LayerNorm + BatchNorm

TABLE VI
PPO HYPERPARAMETERS

Parameter	Value
Learning Rate	3×10^{-4}
Discount Factor (γ)	0.995
GAE Lambda (λ)	0.95
Clip Range	0.2
Batch Size	256
Number of Epochs	8
Entropy Coefficient	0.005

which uses a trained GNN + PPO policy to determine, for each edge, whether to maintain or prune the communication link. These decisions are sent back to the C++ simulation via the same TCP socket. Upon receiving the decision, the simulation enforces the pruning action by suppressing message forwarding between affected vehicles in the next simulation step. This tight integration allows the system to maintain temporal consistency and ensures that detection and control occur continuously during the simulation runtime.

B. Evaluation Datasets

All evaluation results are based on simulation-generated datasets that are completely separate from those used during training and fine-tuning. These test datasets were generated with varied road structures, traffic densities, and attack ratios. This allowed us to assess the generalization and robustness of the full system.

C. GRL Architecture and Training Configuration

The GRL agent was trained offline for 100,000 training steps on a single NVIDIA A100-SXM4-80GB GPU. The training process took approximately 2.5 hours. To ensure reproducibility of the GRL module, we provide a complete description of the GNN architecture and the hyperparameters used to train the PPO agent. Table V summarizes the architecture of the GNN, including the number of layers, hidden dimensions, attention heads, and activation functions. Table VI lists the full PPO training configuration. All parameters were tuned for stability and performance within the simulation domain. Our code is publicly accessible on GitHub ².

D. Evaluation and Ablation Study

In this section, we evaluate the complete VANET security framework and analyze the contribution of the GRL module through ablation to the baseline classifier.

Table VII presents the performance comparison across three representative attack types: Constant Position Offset, Random Speed, and DoS. Tables VIII, IX, and X summarize the performance comparison in three different road configurations, varying vehicle density (20, 50, 100 vehicles) and malicious vehicle ratios (10% to 50%). Each table compares the baseline Informer performance with the proposed Informer+GRL pipeline in terms of precision, recall, F1-score, and accuracy. The improvement column reflects the gain in F1-score, which we consider to be the most representative metric in this setting. Overall, the proposed framework demonstrates consistent performance improvements in all road scenarios and evaluation settings, highlighting the generalizability of our approach. The GRL component improves the reliability of the classification under varying traffic densities and attack intensities. In the following, we summarize the key findings that emerge from this evaluation:

1) *Performance Across Attack Types*: Table VII reports model performance across three attack types under the same traffic scenario. The Constant Position Offset attack shows the lowest baseline F1-score (0.8085), as it preserves realistic speed and acceleration while introducing a persistent spatial displacement that is difficult for time-series models to detect. This is where the GRL agent provides the largest F1-score improvement (+2.36%), as graph-based reasoning exposes spatial inconsistencies between neighboring vehicles.

The DoS Disruptive and Random Speed attacks yield similar baseline F1-scores (≈ 0.828), with smaller GRL F1-score gains (+0.41% and +0.39%, respectively). Random Speed introduces subtle Gaussian noise that is effectively captured by temporal patterns, while DoS primarily affects message volume and congestion, offering limited stable structural cues for graph-based correction.

Across all attack types, Recall gains exceed Precision gains, indicating that GRL mainly reduces false negatives by correcting missed detections, while maintaining a low false-positive rate.

2) Performance by road scenario:

- Road 3 (Table X) shows the most significant optimization potential, with an average improvement in the F1-score of +3.18% and a peak gain of +17.86% in the most challenging setting (20 vehicles, 50% malicious nodes). This indicates that the GRL agent is highly effective in handling complex or irregular environments, where the base Informer model tends to make more classification errors.
- Road 2 (Table IX) achieves the highest absolute performance, with an average F1-score of 83.09% after applying GRL. However, the average improvement is smaller (+1.60%), suggesting that Informer already performed strongly in this structured setting, leaving less room for post-classification correction.
- Road 1 (Table VIII) demonstrates consistent refinement capabilities, with an average improvement of +2.32% in F1-score in all densities and attack levels. This confirms the robustness of the GRL post-processing module in moderately complex network configurations.

3) Impact of Network Density:

²Source code is available at: <https://github.com/latifa1999/VanetSecurity>

TABLE VII
PERFORMANCE COMPARISON ACROSS ATTACK TYPES

Attack Type	Method	Precision	Recall	F1-Score	Improvement (F1-score)
Constant Position Offset	Informer	0.9303	0.7284	0.8085	+2.36%
	Informer+GRL	0.9633	0.7550	0.8276	
DoS Disruptive	Informer	0.9691	0.7473	0.8285	+0.41%
	Informer+GRL	0.9709	0.7524	0.8319	
Random Speed	Informer	0.9680	0.7468	0.8280	+0.39%
	Informer+GRL	0.9706	0.7515	0.8312	

- The Informer+GRL system performs best in low-density scenarios, particularly with 20 vehicles, where it achieves the highest average improvement in F1-score on all roads (up to +5.88%). In such settings, the graph structure is simpler, making malicious behavior more distinguishable through relational context.
- In denser environments (100 vehicles), the average gain drops (e.g., +1.55%), likely due to increased message overlap, graph sparsity and noise, which limit the agent's ability to confidently distinguish anomalous patterns.

4) Effect of Attack Intensity:

- The GRL layer is especially effective under high attack conditions. On Road 3, it achieves a gain of nearly +17% in F1-score at 50% malicious nodes, showcasing its capacity to correct classification errors when the Informer becomes less reliable.
- Even under lower threat levels (e.g., 10–20% malicious), the GRL consistently contributes positive improvements, highlighting its generalization capability across varying degrees of adversarial influence.

5) *Interaction Between Vehicle Density and Attack Intensity:* Tables VIII, IX and X reveal how vehicle density and attack intensity jointly influence GRL effectiveness.

The strongest improvements occur in low-density settings with high attack intensity. At 20 vehicles and 50% malicious nodes, GRL achieves +17.86% on Road 3 and +4.46% on Road 1 in the F1-score. In such cases, the sparse graph provides clearer topological structure, and the higher error rate gives the agent more opportunities to correct classification mistakes.

In contrast, high density with low attack intensity yields minimal gains in F1-score. At 100 vehicles with 10% malicious nodes, the improvements drop to +0.25% (Road 1) and +0.48% (Road 2). Here, the dense graph structure obscures relational patterns, and there are few classification errors to correct. Moreover, the interaction effect varies with road complexity. On Road 3 (most complex), the performance in the F1-score improves steadily with the attack ratio in low-density conditions (+0.93% at 10% to +17.86% at 50%). This suggests that GRL is most helpful when the Informer struggles in challenging environments. However, on simpler roads, improvements tend to level off at moderate attack ratios (20–30%), indicating less room for graph-based correction.

This ablation study confirms that the integration of GRL into the VANET DT significantly enhances the mitigation of anomalies. The GRL agent consistently improves performance over the base classifier, especially in complex road layouts, low-density scenarios, and high-attack conditions.

These results position GRL as a powerful, context-aware post-classification optimization layer capable of maintaining communication integrity in diverse and dynamic VANET environments.

VI. DISCUSSION AND FUTURE WORK

This work presents a complete real-time security framework for VANETs that integrates time-series-based anomaly detection with post-detection mitigation via GRL. Unlike prior approaches that separate detection from response, this system processes V2V message streams continuously, classifies messages based on temporal patterns, and adapts the communication graph by pruning links identified as risky. Experimental results in multiple road topologies and attack intensities show that the framework improves system resilience, especially under sparse traffic and high-threat scenarios, by correcting false positives and false negatives and maintaining message flow integrity. Additionally, the use of a fine-tuning pipeline allows us to leverage large-scale public datasets for general representation learning while efficiently adapting to simulated conditions with far fewer data.

However, several important aspects deserve further discussion. The proposed GRL-based framework demonstrates consistent performance across varying traffic densities (20, 50, and 100 vehicles), indicating its ability to scale to moderately sized VANET scenarios. As the number of vehicles increases, the communication graph becomes larger and more complex, leading to higher computational and memory requirements for real-time decision-making. While the current centralized design remains effective within the evaluated settings, future work will explore Multi-Agent Graph Reinforcement Learning (MAGRL) to enable distributed decision-making over localized subgraphs, improving scalability and responsiveness in large-scale environments.

In addition, although the framework is evaluated on representative attack types (DoS, Random Speed, and Position Offset), its design is general and can be extended to incorporate a broader range of attack models. This includes more sophisticated threats such as Sybil attacks, replay attacks, and coordinated multi-node misbehavior, which can be integrated within the same detection-mitigation pipeline.

From a deployment perspective, the proposed framework is designed to operate in real time within a Digital Twin environment; however, its practical implementation would require integration with edge computing infrastructure such as Road Side Units (RSUs). Challenges such as communication latency, synchronization between simulation and real-world

TABLE VIII
PERFORMANCE ANALYSIS ON ROAD 1: INFORMER VS INFORMER + GRL ACROSS DIFFERENT VEHICLE DENSITIES AND ATTACK INTENSITIES

Vehicles	Attack %	Method	Performance Metrics (%)				Improvement (F1-score)
			Precision	Recall	F1-Score	Accuracy	
20	10%	Informer	89.11	74.51	79.46	74.51	+2.12%
		Informer+GRL	93.28	76.93	81.58	76.93	
	20%	Informer	81.11	75.88	77.16	75.88	+5.36%
		Informer+GRL	89.18	81.39	82.52	81.39	
	30%	Informer	84.44	78.70	80.44	78.70	+0.82%
		Informer+GRL	90.09	79.06	81.26	79.06	
50%	Informer	80.91	80.92	80.92	80.92	+4.46%	
	Informer+GRL	89.15	85.35	85.38	85.35		
50	10%	Informer	83.72	77.93	79.96	77.93	+1.27%
		Informer+GRL	89.89	79.17	81.23	79.17	
	20%	Informer	81.00	79.94	80.20	79.94	+3.37%
		Informer+GRL	88.54	83.27	83.57	83.27	
	30%	Informer	84.23	78.40	80.25	78.40	+1.18%
		Informer+GRL	90.38	79.09	81.43	79.09	
50%	Informer	88.81	74.57	78.97	74.57	+1.98%	
	Informer+GRL	92.39	76.83	80.95	76.83		
100	10%	Informer	87.01	78.06	80.88	78.06	+0.25%
		Informer+GRL	90.57	78.54	81.13	78.54	
	20%	Informer	83.95	75.90	78.45	75.90	+1.05%
		Informer+GRL	90.33	76.54	79.50	76.54	
	30%	Informer	87.21	76.66	79.97	76.66	+0.73%
		Informer+GRL	91.65	77.11	80.70	77.11	
50%	Informer	86.60	71.93	77.26	71.93	+1.28%	
	Informer+GRL	92.86	73.07	78.54	73.07		
Informer Average			86.49	77.19	79.51	77.19	+2.32%
Informer+GRL Average			90.45	79.31	81.83	79.31	

TABLE IX
PERFORMANCE ANALYSIS ON ROAD 2: INFORMER VS INFORMER+GRL ACROSS DIFFERENT VEHICLE DENSITIES AND ATTACK INTENSITIES

Vehicles	Attack %	Method	Performance Metrics (%)				Improvement (F1-score)
			Precision	Recall	F1-Score	Accuracy	
20	10%	Informer	81.92	78.26	78.77	78.26	+7.55%
		Informer+GRL	90.01	85.99	86.32	85.99	
	20%	Informer	82.28	81.86	81.73	81.86	+6.78%
		Informer+GRL	90.72	88.73	88.51	88.73	
	30%	Informer	82.81	82.69	82.72	82.69	+2.90%
		Informer+GRL	89.15	85.71	85.62	85.71	
50%	Informer	88.90	87.80	87.79	87.80	+1.25%	
	Informer+GRL	91.05	89.26	89.04	89.26		
50	10%	Informer	89.11	76.65	80.77	76.65	+1.04%
		Informer+GRL	92.91	77.59	81.81	77.59	
	20%	Informer	83.30	80.48	81.40	80.48	+0.92%
		Informer+GRL	89.70	80.76	82.32	80.76	
	30%	Informer	83.73	76.64	78.77	76.64	+2.03%
		GRL	90.10	78.45	80.80	78.45	
50%	Informer	88.97	76.12	80.15	76.12	+1.04%	
	Informer+GRL	92.43	77.13	81.19	77.13		
100	10%	Informer	82.48	82.06	82.21	82.06	+0.48%
		Informer+GRL	88.38	82.19	82.69	82.19	
	20%	Informer	83.83	83.54	83.63	83.54	+0.53%
		Informer+GRL	88.67	83.99	84.16	83.99	
	30%	Informer	91.45	74.98	80.41	74.98	+2.07%
		Informer+GRL	93.97	77.59	82.48	77.59	
50%	Informer	92.11	74.78	81.01	74.78	+0.34%	
	Informer+GRL	94.86	75.06	81.35	75.06		
Informer Average			85.26	79.89	81.49	79.89	+1.60%
Informer+GRL Average			88.63	81.41	83.09	81.41	

data, and distributed coordination must be carefully addressed to ensure reliable operation in real-world scenarios.

Future work will further investigate distributed GRL strategies and enhance robustness against adaptive and evolving attack behaviors, enabling the system to maintain reliable performance in increasingly complex VANET environments.

VII. CONCLUSION

Securing VANETs against malicious behaviors remains a fundamental challenge due to their dynamic topology, real-time communication demands, and susceptibility to data-level attacks. Traditional detection methods often rely on static features or fail to address how the system should respond after

TABLE X
PERFORMANCE ANALYSIS ON ROAD 3: INFORMER VS INFORMER + GRL ACROSS DIFFERENT VEHICLE DENSITIES AND ATTACK INTENSITIES

Vehicles	Attack %	Method	Performance Metrics (%)				Improvement (F1-score)
			Precision	Recall	F1-Score	Accuracy	
20	10%	Informer	86.40	78.15	80.46	78.15	+0.93%
		Informer+GRL	90.60	78.87	81.39	78.87	
	20%	Informer	75.11	70.88	72.04	70.88	+10.72%
		Informer+GRL	88.90	81.91	82.76	81.91	
	30%	Informer	73.77	72.72	73.02	72.72	+9.61%
		Informer+GRL	88.00	82.42	82.63	82.42	
50%	Informer	67.47	67.52	67.49	67.52	+17.86%	
	Informer+GRL	88.96	85.50	85.35	85.50		
50	10%	Informer	82.85	79.07	80.09	79.07	+0.66%
		Informer+GRL	88.66	79.41	80.75	79.41	
	20%	Informer	82.11	78.54	79.71	78.54	+1.35%
		Informer+GRL	89.42	79.29	81.06	79.29	
	30%	Informer	80.26	77.68	78.53	77.68	+3.69%
		Informer+GRL	89.14	81.01	82.22	81.01	
50%	Informer	80.89	76.29	77.78	76.29	+3.74%	
	Informer+GRL	89.62	79.76	81.52	79.76		
100	10%	Informer	86.45	72.07	77.20	72.07	+4.04%
		Informer+GRL	92.93	76.77	81.24	76.77	
	20%	Informer	91.14	79.03	83.31	79.03	+6.33%
		Informer+GRL	93.71	86.00	89.69	86.00	
	30%	Informer	88.51	76.19	80.47	76.19	+0.18%
		Informer+GRL	92.13	76.64	80.65	76.64	
50%	Informer	85.35	77.72	79.87	77.72	+1.30%	
	Informer+GRL	90.26	78.82	81.17	78.82		
Informer Average			81.62	76.54	78.36	76.54	+3.18%
Informer+GRL Average			86.70	80.49	81.54	80.49	

detecting suspicious behavior, especially in the presence of FP or FN.

This work presents a complete real-time VANET security framework that integrates time-series-based anomaly detection with post-detection control through GRL. By fine-tuning a state-of-the-art temporal model on V2V communication data, the system is capable of detecting anomalies at the message level. The GRL agent then interprets these outputs in the context of a dynamic communication graph and learns to prune or retain links to contain the spread of malicious information and improve network reliability.

Through simulation-based evaluation across varying road scenarios and attack conditions, the system demonstrated improved resilience and adaptability, particularly in difficult cases such as sparse traffic or high attack ratios. An ablation study further confirmed the added value of GRL in enhancing detection robustness and correcting classification errors in real time.

Overall, this work contributes to a unified and data-oriented security architecture that brings together detection and control in a single loop. It provides a foundation for building future intelligent and adaptive security systems capable of learning, reacting, and adapting to evolving threats in VANET environments.

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